
Redyx — Community Service Test Cases

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Community Service — Test Case Specification

Project Name	Redyx
Module Name	Community
File	internal/community/server_test.go
Total Tests	16
Package	community (same-package test)
Created By	idityaGE
Date of Creation	2026-05-10

Test Infrastructure

Component	Implementation
Logger	<code>zap.NewNop()</code>
DB Pool	<code>nil</code> — tests short-circuit before DB access
Cache	<code>nil</code> — not required for validation tests
Auth Context	<code>auth.WithClaims(context.Background(), &auth.Claims{...})</code>
Helper convention	<code>t.Helper()</code> in <code>testCommunityServer</code> , <code>authedCommCtx</code>
Nil recovery	<code>defer recover()</code> for tests that pass validation and reach nil DB

1. nameRegex Tests

Test Case ID	Test Name	Pre-Condition	Test Steps	Test Data	Expected Result	Status
TC_COMM_001	TestNameRegex_ValidNames	nameRegex compiled	Call <code>nameRegex.MatchString</code> for 6 valid names	"abc", "Göteborg", "community123", "a_b_c_d_e_f", "ALLCAPS", "Mix3dC4se"	All 6 names match the regex	PASS

Test Case		Pre-	Test Steps	Test Data	Expected Result	Status
ID	Test Name	Condition				
TC_ COMM_ 002	TestNameRegex_InvalidNames	nameRegex compiled	Call <code>nameRegex.Match()</code> for 6 invalid names	"ab" (too short), "" (empty), "has space", "has-dash", "has.dot", "toolong-name123456789" (22 chars, over limit)	None of the 6 names match the regex	PASS

2. CreateCommunity Validation Tests

Test Case		Pre-	Test Steps	Test Data	Expected Result	Status
ID	Test Name	Condition				
TC_ COMM_ 003	TestCreateCommunity_Unauthenticated	Only community server initialized	Call <code>CreateCommunity</code> with unauthenticated context	name = "validname"	Returns an error (missing auth claims)	PASS
TC_ COMM_ 004	TestCreateCommunity_InvalidName_TooShort	Authenticated context	Call <code>CreateCommunity</code> with 2-character name	name = "ab" (min is 3)	Returns an error	PASS
TC_ COMM_ 005	TestCreateCommunity_InvalidName_WithSpace	Authenticated context	Call <code>CreateCommunity</code> with name containing space	name = "my community"	Returns an error	PASS
TC_ COMM_ 006	TestCreateCommunity_InvalidName_WithDash	Authenticated context	Call <code>CreateCommunity</code> with name containing dash	name = "my-community"	Returns an error	PASS
TC_ COMM_ 007	TestCreateCommunity_InvalidName_TooLong	Authenticated context	Call <code>CreateCommunity</code> with 23-character name	name = "toolong-name123456789012" (23 chars, max is 21)	Returns an error	PASS
TC_ COMM_ 008	TestCreateCommunity_ValidName_FailsAtDB	Authenticated context; nil DB pool	Call <code>CreateCommunity</code> with valid name; recover panic	name = "validname", description = "A test community"	Validation passes; panics at nil DB (recovered) or returns DB error	PASS

3. UpdateCommunity Validation Tests

Test Case ID	Test Name	Pre-Condition	Test Steps	Test Data	Expected Result	Status
TC_COMM_009	TestUpdateCommunityUnauthenticated	Community server initialized	Call UpdateCommunity with unauthenticated context	name = "validname", description = "New description"	Returns an error	PASS

4. Auth Guard Tests (JoinCommunity / LeaveCommunity / AssignModerator / RevokeModerator)

Test Case ID	Test Name	Pre-Condition	Test Steps	Test Data	Expected Result	Status
TC_COMM_010	TestJoinCommunityUnauthenticated	Community server initialized	Call JoinCommunity with unauthenticated context	name = "test-community"	Returns an error	PASS
TC_COMM_011	TestLeaveCommunityUnauthenticated	Community server initialized	Call LeaveCommunity with unauthenticated context	name = "test-community"	Returns an error	PASS
TC_COMM_012	TestAssignModeratorUnauthenticated	Community server initialized	Call AssignModerator with unauthenticated context	name = "test-community", userId = "user-2", username = "otheruser"	Returns an error	PASS
TC_COMM_013	TestRevokeModeratorUnauthenticated	Community server initialized	Call RevokeModerator with unauthenticated context	name = "test-community", userId = "user-2"	Returns an error	PASS

5. communityRulesToSlice Tests

Test Case ID	Test Name	Pre-Condition	Test Steps	Test Data	Expected Result	Status
TC_COMM_014	TestCommunityRulesToSlice_Call communityRulesToSlice ConvertsCorrectly		Call communityRulesToSlice with 2 rules	rules = [{title:"Be Kind", desc:"Treat others well"}, {title:"No Spam", desc:"No promotional content"}]	Returns slice of 2 maps; result[0]["title"]="Be Kind", result[1]["title"]="No Spam"	PASS
TC_COMM_015	TestCommunityRulesToSlice_Call communityRulesToSlice Empty		Call communityRulesToSlice with empty slice	rules = []	Returns empty slice (length 0)	PASS
TC_COMM_016	TestCommunityRulesToSlice_Call communityRulesToSlice Nil		Call communityRulesToSlice with nil	rules = nil	Returns empty slice (length 0)	PASS

6. buildCommunityProto Tests

Test Case ID	Test Name	Pre-Condition	Test Steps	Test Data	Expected Result	Status
TC_COMM_017	TestBuildCommunityProto_NoRules		Call buildCommunityProto with nil rules	communityId = "comm-id-1", name = "testcomm", memberCount = 42, visibility = int16(1)	No error; CommunityId="comm-id-1", Name="testcomm", MemberCount=42, Rules is empty	PASS
TC_COMM_018	TestBuildCommunityProto_WithRules	Valid JSON rules	Call buildCommunityProto with valid JSON rules	rulesJSON = [{"title":"Rule 1","description":"Desc 1"}, {"title":"Rule 2","description":"Desc 2"}]	No error; len(Rules)=2, Rules[0].Title "1"	PASS

Test Case ID	Test Name	Pre-Condition	Test Steps	Test Data	Expected Result	Status
TC_COMM_019	TestBuildCommunityProto_In-validRulesJSON		Call buildCommunityProto with invalid JSON bytes	rulesJSON = "not valid json"	Returns an error (JSON unmarshal fails)	PASS

Patterns & Conventions

Pattern	Implementation
Package scope	package community (white-box / same-package)
Logger	zap.NewNop() (silent)
Nil DB/cache	NewServer(nil, nil, zap.NewNop()) — validation and helper tests only
Panic recovery	defer recover() guards for tests that reach nil DB
Auth context	auth.WithClaims(context.Background(), &auth.Claims{UserID, Username})
Helper tagging	t.Helper() in testCommunityServer and authedCommCtx
Name rules	Regex enforces 3–21 chars, alphanumeric + underscore only